

Model ID	Statistical Method	Model Formula	Analytical Purpose
RF-FIELD	Random Forest ¹	Predicted_WL ~ 19 immune genes (field application)	Generate immune-based
PL-1	parasiteLoad ²	response ~ HI + hHe + $\alpha \times hHe$ Sex ³	Test sex-specific hybrid c
PL-2	parasiteLoad ²	response ~ HI + hHe + $\alpha \times hHe$ Sex (uninfected) ³	Test constitutive hybrid c
PL-3	parasiteLoad ²	response ~ HI + hHe + $\alpha \times hHe$ Infection_status ³	Test infection dominance
LM-1	Linear Regression	Predicted_WL ~ HI $\times$ hHe	Test genetic baseline effe
LM-2	Linear Regression	Predicted_WL ~ Sex + HI + hHe + Infection	Test main effects model
LM-3	Linear Regression	Predicted_WL ~ Sex $\times$ HI + interactions	Test interaction effects

Bold = $p < 0.05$

¹ Random Forest model from Webster et al. (in review)

² parasiteLoad package implements Baird et al. (2012) framework

³ α represents deviation from additivity (positive = increased costs with M. m. musculus ancestry)

